# The ML Lifecycle

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

---



## Recall and choose your starting point

> Record your starting explanation for The ML Lifecycle before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a **label**led second line to correct your explanation.

### Creator Defense

Without opening the old lesson, explain one key idea from Creator Defense. Give an example and a mistake that the idea helps you avoid.

**My remembered idea, example and correction**

---

## Practise with fading help

> Complete the supported The ML Lifecycle trace and record the first step you can explain without help.

Use W2: Trace one candidate for The **machine learning** lifecycle. Record the answer once; the lesson and workbook refer to the same attempt.

| Given inputs | Procedure I chose | Middle steps | Result and check |
|---|---|---|---|
| | | | |

> **The step I completed without help, and the hint I needed**

---

WORKBOOK · 3 OF 21

## Individual reasoning check

> Check the proposed The ML Lifecycle explanation against the supplied facts before discussing it.

Use the worked case for The **machine learning** lifecycle. Proposed explanation: “The model wrote its own task”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

**My explanation before partner discussion**

---

WORKBOOK · 4 OF 21

## Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

### Independent case: Fit and evaluate independently

New **training** cards are (width 2, **label** 0), (4, 0), (6, 1), (8, 1). Candidates are 3, 5 and 7. Predict 1 when width is at least the threshold.

Count **training** mistakes, select the fewest with smaller threshold breaking ties, and record the choice before testing. Held out cards are (3, 0), (7, 1), (5, 0).

Give each test **prediction**, the correct count and a limited conclusion.

**My independent explanation, trace and limits**

---

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

---

WORKBOOK · 5 OF 21

## Your learning target

> Connect the The ML Lifecycle target to a trace and a justified conclusion.

A team **label**s photos, fits a model and celebrates one correct new photo. List the missing **evaluation** steps and define the intended use before testing.

---

WORKBOOK · 6 OF 21

## Solve the given case

> Solve the supplied The ML Lifecycle case and show the deciding evidence.

A team **label**s leaf photos, trains a healthy/spotted classifier and shows one correct **prediction**.

Order the lifecycle and identify missing **evidence**.

Show your trace, calculation or **evidence**

> **Write or draw your reasoning and evidence.**

---

WORKBOOK · 7 OF 21

## Check a changed case

> Apply the procedure to the changed case and record why the conclusion follows.

A team selects a model before deciding what outcome to predict or where the data comes from. Name two missing lifecycle decisions and their effect.

My independent answer and reason

> **Write or draw your reasoning and evidence.**

**Week 1 Test and reflect**

---

WORKBOOK · 8 OF 21

## Design a revealing test

> Specify a changed input, expected result and observation separately.

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

| Test record | My evidence |
| --- | --- |
| Changed input or condition | |
| Expected answer or fallback | |
| Observed result | |
| Why the result occurred | |

---

WORKBOOK · 9 OF 21

## Project checkpoint

> Justify what The ML Lifecycle adds to Responsible Plant Classifier and keep the evidence.

Write the Responsible Plant Classifier problem statement, intended user, **prediction** target, success measure, and human-review point.

The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

---



## Self check

> Judge explanation, evidence, transfer and limits separately; attach the work behind your judgement.

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

WORKBOOK · 11 OF 21

## Transfer practice from the original programme

> Use this earlier case as transfer practice. A case already discussed with feedback is not fresh assessment evidence.

A team **label**s photos, fits a model and celebrates one correct new photo. List the missing **evaluation** steps and define the intended use before testing.

> **My independently reasoned answer**

Record support used: none, hint or teacher explanation.

---

WORKBOOK · 12 OF 21

## W1 Identify the learning decision

> Identify the learning decision. Apply the supplied The ML Lifecycle facts and keep your reasoning.

For the worked cards, name the **input**, **label**, three candidate thresholds and the information allowed to select a threshold.

**My reasoning and evidence**

---

WORKBOOK · 13 OF 21

## W2 Trace one candidate

> Trace one candidate. Apply the supplied The ML Lifecycle facts and keep your reasoning.

Use threshold 6 on the four **training** widths 1, 3, 5 and 7. Write all **prediction**s, identify each error, and explain why threshold 4 is selected.

**My reasoning and evidence**

---

WORKBOOK · 14 OF 21

## W3 Repair the report

> Repair the report. Apply the supplied The ML Lifecycle facts and keep your reasoning.

A learner fits on all six worked cards and then reports 6 of 6 as independent test performance. Explain what is wrong and what **evidence** is needed.

**My reasoning and evidence**

WORKBOOK · 15 OF 21

## W4 Fit and evaluate independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Fit and evaluate independently. Apply the supplied The ML Lifecycle facts and keep your reasoning.

New **training** cards are (width 2, **label** 0), (4, 0), (6, 1), (8, 1). Candidates are 3, 5 and 7. Predict 1 when width is at least the threshold.

Count **training** mistakes, select the fewest with smaller threshold breaking ties, and record the choice before testing. Held out cards are (3, 0), (7, 1), (5, 0).

Give each test **prediction**, the correct count and a limited conclusion.

**My reasoning and evidence**

WORKBOOK · 16 OF 21

## W5 Decide the next step

> Decide the next step. Apply the supplied The ML Lifecycle facts and keep your reasoning.

After W4, a team changes its threshold because of the test error. Can the same three cards be an untouched final test for the changed model? Explain how to proceed.

**My reasoning and evidence**

---



## Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

| Evidence | My record |
|---|---|
| Worked alone / hint / model | |
| First step I repaired and why | |
| New case checked later; date and result | |
| What I can now explain without notes | |
| One remaining question | |

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

# Pictorial case practice: The ML Lifecycle

> Follow one complete plant-classifier lifecycle without tuning on the final test. Explain the labelled step before checking the changed condition.

Follow one complete plant-classifier lifecycle without tuning on the final test.

Toy **training** leaves have widths 1 healthy, 3 healthy, 5 spotted, 7 spotted. Candidates predict spotted at width>=2 or >=4.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-4/1

# The ML Lifecycle

**Start with the supplied facts**

Follow each arrow. Ask which fact allows the next step.



### 1. Fit using training

Cutoff 2 makes one error at width 3; cutoff 4 makes zero training errors.





### 2. Freeze candidate

Choose cutoff 4 under the stated training-error rule.





### 3. Test once

Fresh leaves width 2 healthy and 6 spotted are both predicted correctly: 2/2.

**CHANGE ONE THING**

The fresh width-6 leaf is actually healthy.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Should the team silently retune on the changed test? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

**My picture-based explanation and changed-case prediction**

### Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

**Label** this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

---



## Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Responsible Plant Classifier record.

Project: Responsible Plant Classifier

Current milestone: Frame the need. Name the user, the problem, allowed **input**s and a safe simple starting method.

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

**Label** this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

| Evidence to keep | My record |
|---|---|
| This week's input and deciding step | |
| Prediction and actual result (or labelled paper trace) | |
| One change since the previous checkpoint and why | |

| Evidence to keep | My record |
| --- | --- |
| One error or limit and the next test | |

**Explain what this week adds to your project. Reuse last week's record rather than starting again.**

WORKBOOK · 20 OF 21

## Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Follow one complete plant-classifier lifecycle without tuning on the final test.

### Practice 1: explain the deciding evidence

Which **label**led step matches this detail? Fresh leaves width 2 healthy and 6 spotted are both predicted correctly: 2/2.

1. Test once

2. Freeze candidate

3. Fit using **training**

**My selected conclusion, deciding fact and explanation**

### Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

### Feedback · compare after trying

After your attempt, compare with the supported explanation: Test once Fresh leaves width 2 healthy and 6 spotted are both predicted correctly: 2/2. Read the picture in order.

Fresh leaves width 2 healthy and 6 spotted are both predicted correctly: 2/2. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

**Practice 2: explain the deciding evidence**

### Changed case · predict first

Changed-case practice: The fresh width-6 leaf is actually healthy. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed **input** or the procedure.

2. This one example guarantees correct results for every future **input**.

3. The frozen model is correct on 1/2 test leaves.

**My selected conclusion, deciding fact and explanation**

### Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

### Feedback · compare after trying

After your attempt, compare with the supported explanation: The frozen model is correct on 1/2 test leaves. The supplied reference changed; moving the cutoff after seeing this test would reuse final **evidence** for development.

### Repair a missing building block

Review Creator Defense (ai-3/36). Goal: Explain the **mechanism** and terminology of generative systems defense. Short practice: Which numbers would you display to make the defense honest?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

### Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, **label** your result as a trace rather than an execution.

### Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

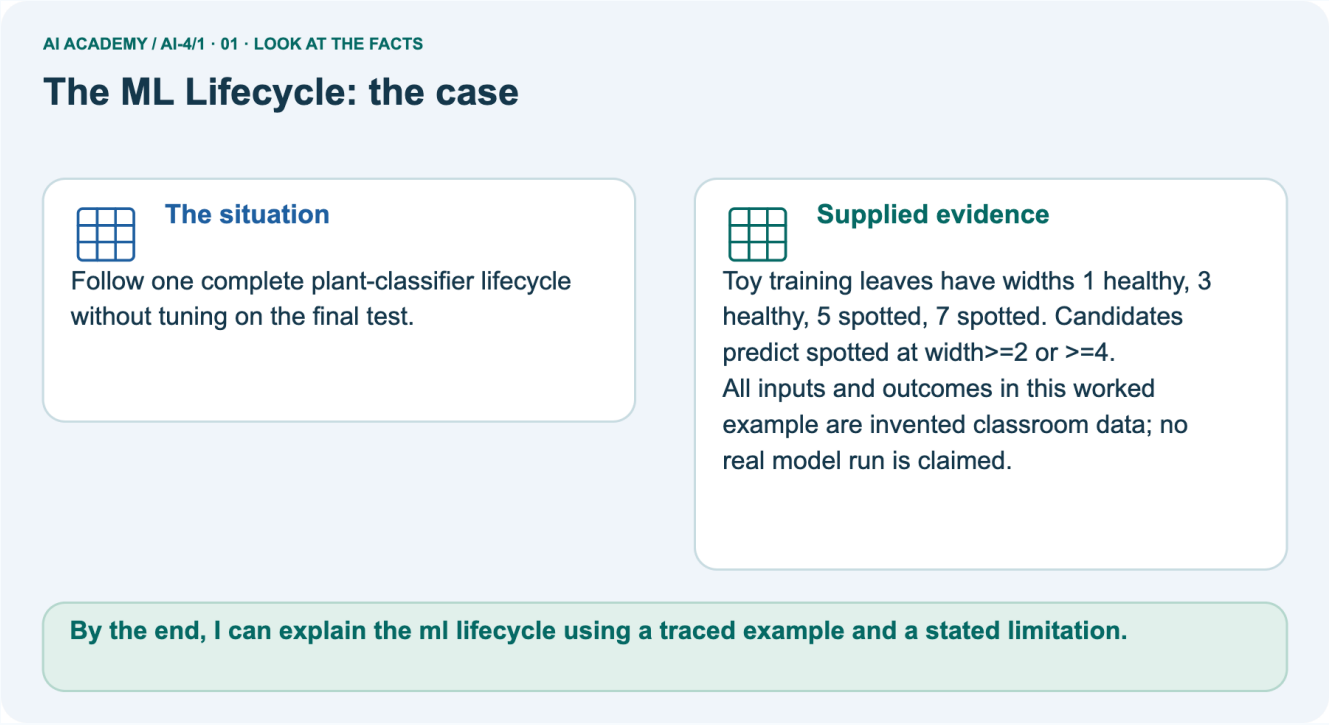
Drafts stay in this browser when storage is available; clear drafts on a shared device.

**What I repaired, the evidence I kept and the next test**

## Visual concept journey

> Explain each The ML Lifecycle picture from its labels, then record a relevant limit.

### 01. The ML Lifecycle: the case




Follow the labelled picture with the matching explanation. Open the vector diagram

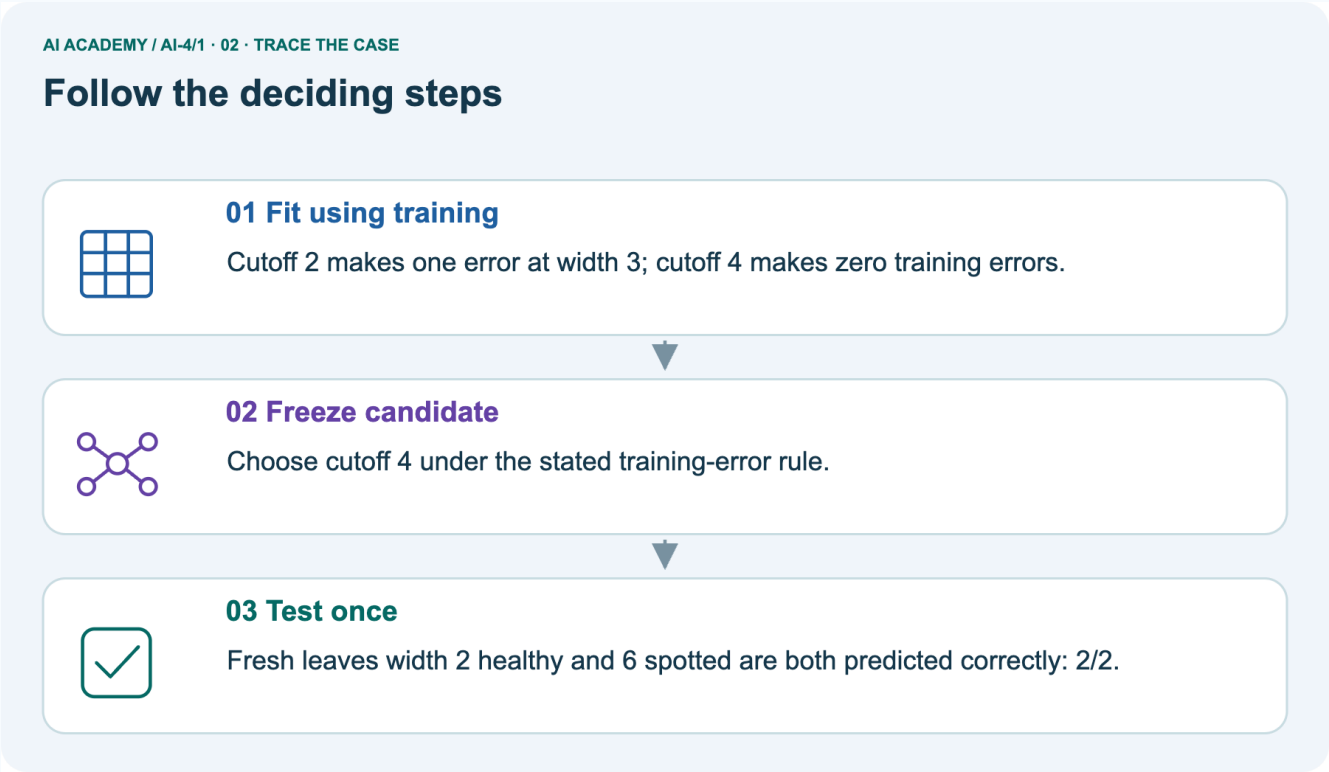
Follow one complete plant-classifier lifecycle without tuning on the final test. Supplied facts: Toy **training** leaves have widths 1 healthy, 3 healthy, 5 spotted, 7 spotted. Candidates predict spotted at width>=2 or >=4.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

> Point to the given fact that matters most. Explain why it matters.

**My picture explanation, deciding evidence and one limit**

## 02. Follow the deciding steps



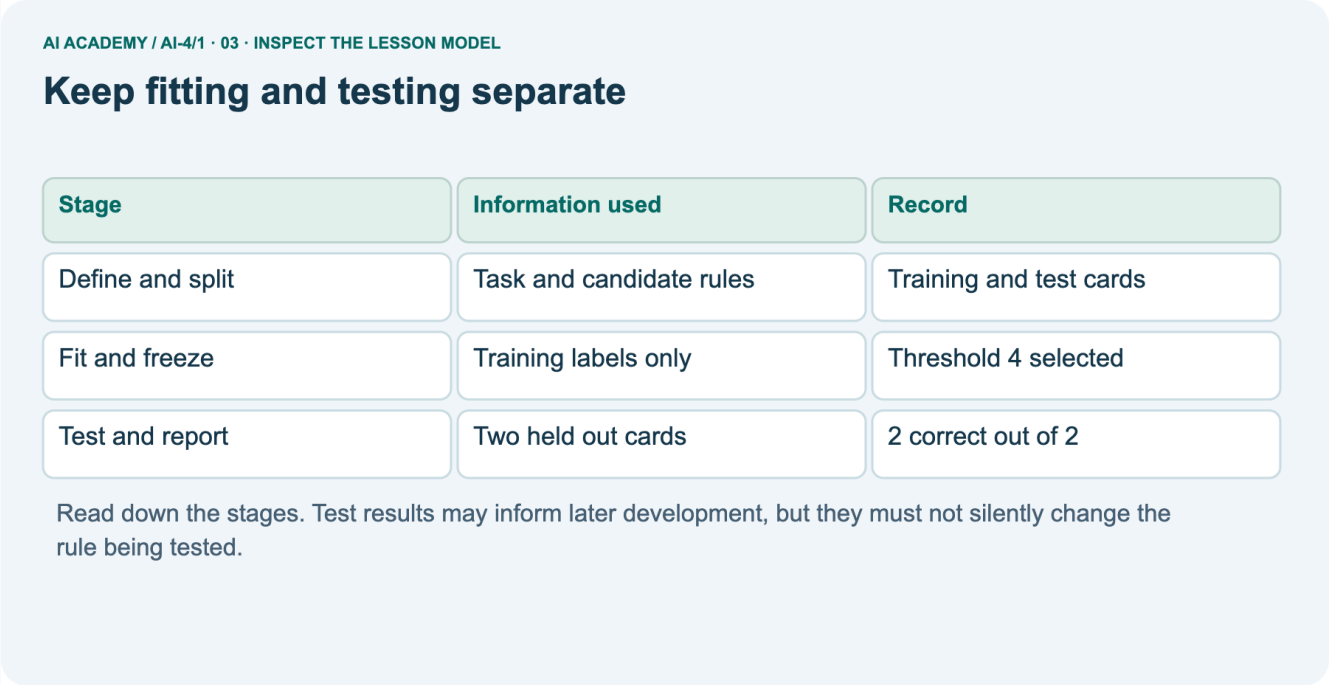

Follow the labelled picture with the matching explanation. Open the vector diagram

Fit using **training**: Cutoff 2 makes one error at width 3; cutoff 4 makes zero **training** errors. → Freeze candidate: Choose cutoff 4 under the stated **training**-error rule. → Test once: Fresh leaves width 2 healthy and 6 spotted are both predicted correctly: 2/2.

Point to each numbered stage and explain which recorded **input** it uses.

**My picture explanation, deciding evidence and one limit**

### 03. Keep fitting and testing separate



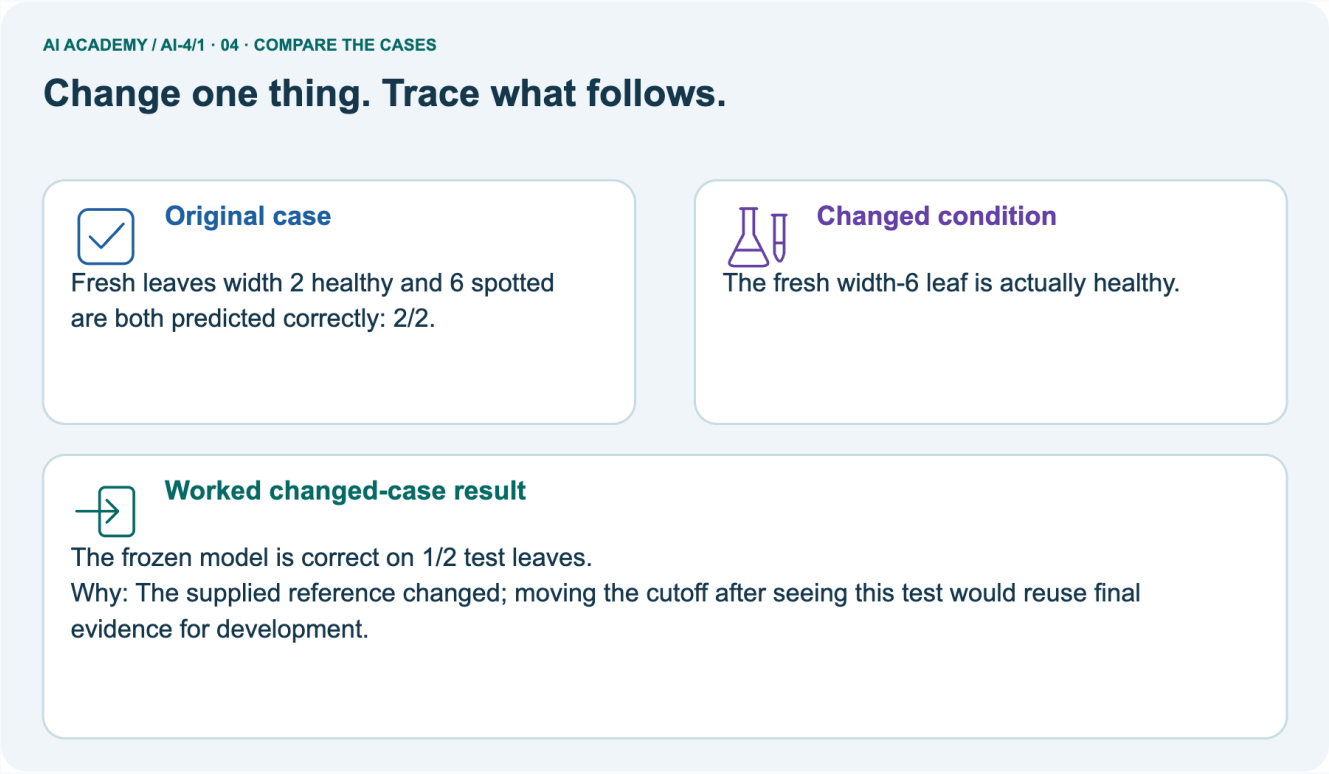

Follow the labelled picture with the matching explanation. Open the vector diagram

Keep **fitting** and testing separate. Define and split | Task and candidate rules | **Training** and test cards. Fit and freeze | **Training label**s only | Threshold 4 selected. Test and report | Two held out cards | 2 correct out of 2

Read the **label**s and values. Explain the relationship the model shows.

**My picture explanation, deciding evidence and one limit**

## 04. Change one thing. Trace what follows.




Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Fresh leaves width 2 healthy and 6 spotted are both predicted correctly: 2/2. Changed condition: The fresh width-6 leaf is actually healthy. Expected result: The frozen model is correct on 1/2 test leaves.

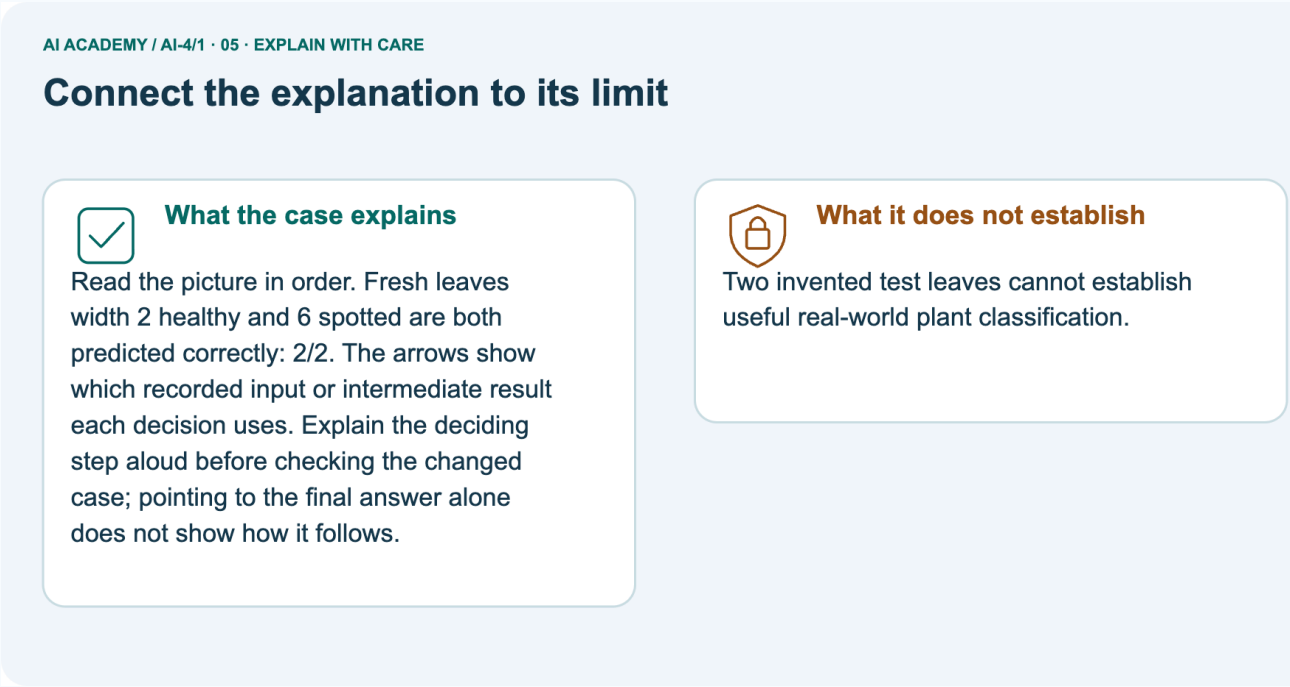
Reason: The supplied reference changed; moving the cutoff after seeing this test would reuse final **evidence** for development.

Predict before checking the worked changed-case result. Name what changed and what the

supplied facts allow you to conclude.

**My picture explanation, deciding evidence and one limit**

### 05. Connect the explanation to its limit




Follow the labelled picture with the matching explanation. Open the vector diagram

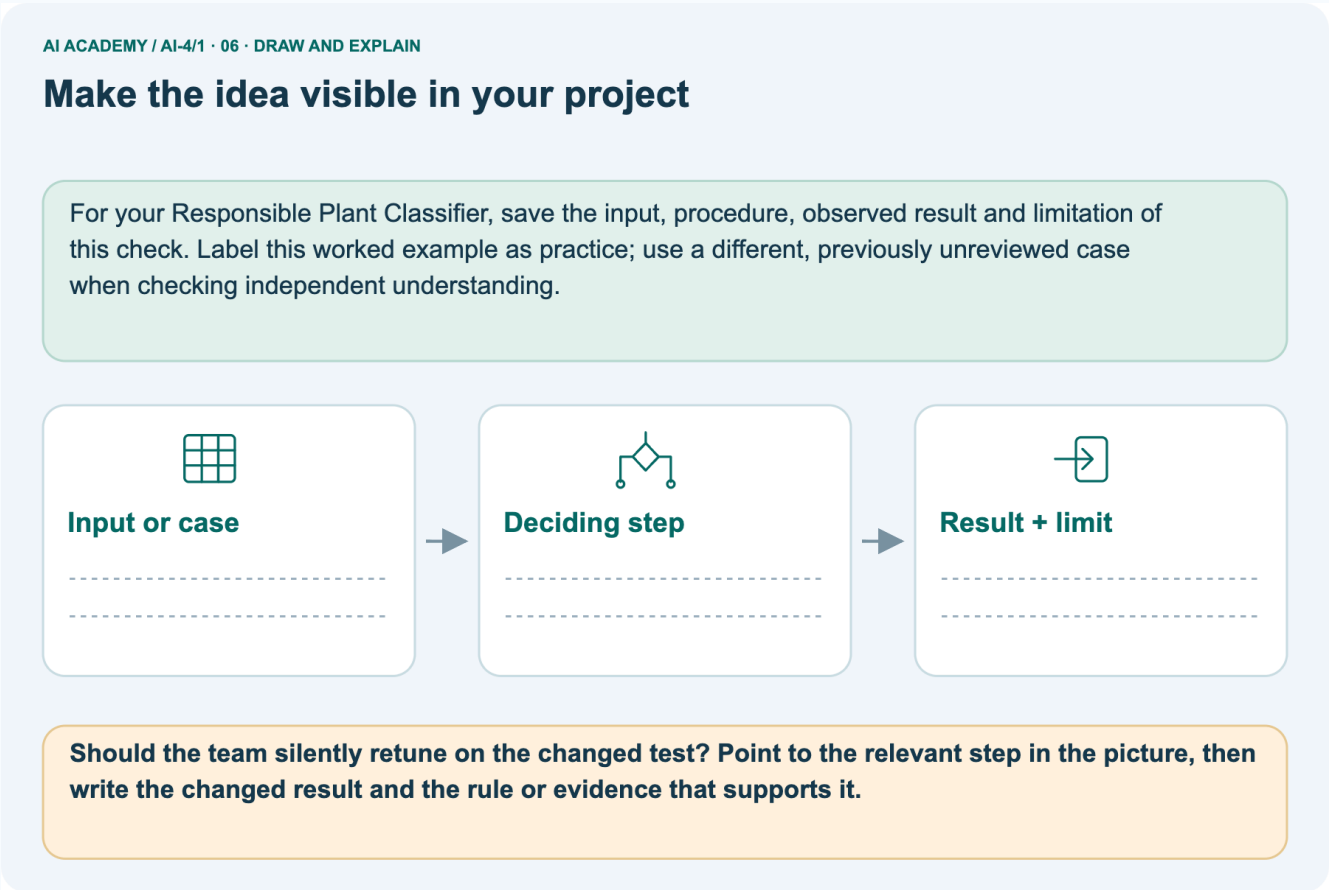
Read the picture in order. Fresh leaves width 2 healthy and 6 spotted are both predicted correctly: 2/2. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows. Limit: Two invented test leaves cannot establish useful real-world plant classification.

Explain the case, then point to a limit on the claim.

**My picture explanation, deciding evidence and one limit**

## 06. Make the idea visible in your project




Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

**Label** this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Should the team silently retune on the changed test? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> Draw your project trace and justify the deciding step using **evidence**.

### My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.